

Red Team AI Malware Simulation Report

Target: target_{name}

Generated: generated_{at}

Simulation: $\text{sim}_{start} \rightarrow \text{sim}_{end}$

Version: Red Team AI Engine v2025.09.11

Author / Team: Red Team / Security Assessment Team

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Executive Summary

Purpose of Simulation:

This simulation assessed the security posture of the target system and tested its defenses against a controlled AI-driven malware attack. The goal was to evaluate how the system responds to various automated offensive techniques – reconnaissance, and attack decision-making.

Key Findings:

- **Open Ports / Services of Concern:**

- p
- N/A

- **Sensitive Files or Data Identified:**

- f
- None

- **Active AV / EDR Detected:**

- av
- Unknown

- **Overall Risk Assessment:** `exec_summary.overall_risk|default("Low")`

AI Model Performance:

- **Recon Prioritisation AI:** `exec_summary.recon_ai|default(0)%|Correctly prioritized exec_summary.recon`

Scope & Methodology

Overview

This engagement is a controlled, simulated assessment executed by the RAPTOR Red Team AI Malware Simulator. The purpose is to evaluate detection, prioritisation, and response capabilities of the target system in a safe research environment — not to perform real-world exploitation. All activity is constrained to authorized test systems and simulated payloads.

Phases

Phase 1 — Reconnaissance (Passive & Active Collection)

Objective: gather an inventory of the target system (OS, services, users, exposed ports, environment variables, and candidate sensitive files) to produce prioritized reconnaissance findings for later phases. Scope: passive discovery first (public-facing services, metadata), followed by limited, consented active scanning in a lab-like environment. Deliverables: structured recon dataset, open port list, environment snapshot, and prioritized asset list.

Phase 2 — C2 Simulation Dummy Actions (Controlled Emulation)

Objective: emulate attacker post-compromise behaviour at a high level — command-and-control interaction, staged dummy actions, and controlled data movement — while avoiding any real malicious payloads or persistence mechanisms. Scope: simulated C2 exchanges and benign stand-in payloads that exercise detection logic and telemetry generation only.

Phase 3 — Reporting & Analysis

Objective: consolidate data from prior phases into an evidence-backed report, quantify model performance, and provide mitigations and recommendations. Deliverables: structured findings, charts, per-module analysis, and an executive summary suitable for stakeholders.

AI Models and Roles

Recon Prioritisation AI (Server-side, Post-processing)

Role: ingest raw reconnaissance artifacts (service lists, file discovery, logs) and rank findings by operational relevance and risk. It helps the analyst focus on high-value assets and probable attack vectors. Outputs: prioritized recon list, confidence scores, and suggested follow-ups.

Attack Decision AI (Server-side, Planner)

Role: select the next benign/simulated action based on recon priorities and mission ob-

jectives. It balances likelihood of success, detection risk, and mission value to recommend the best next simulated step. Outputs: ordered action plans and rationale metadata (e.g., why an action was chosen).

Data, Metrics & Evaluation

Key metrics captured during simulations include:

- **Detection Events:** timestamped telemetry and alert counts for each simulated action.
- **Recon Prioritisation Accuracy:** analyst-validated relevance of top-N findings.
- **Decision Optimality:** proportion of selected actions that met mission objectives under risk constraints.

Evaluation approach: metrics are aggregated per-simulation and across runs; charts (e.g., pie/bar) visualise distributions. All model outputs are accompanied by confidence values and human-readable explanations to aid analyst review.

Safety, Ethics & Operational Controls

This simulation adheres to strict safety and ethical controls:

- **Authorization:** tests are executed only on systems explicitly authorized for simulation. No unsanctioned scanning or exploitation is performed.
- **No Real Exploits:** payloads are inert or simulated placeholders; persistence, credential theft, or destructive actions are not performed.
- **Logging & Isolation:** all activity is logged, contained within isolated test networks, and retained for post-test analysis.
- **Human Oversight:** model decisions and high-risk actions require human review before enactment in any semi-automated workflow.
- **Compliance:** users of the software must comply with applicable laws, institutional policies, and the RAPTOR license terms.

Tools, Techniques & Limitations

Additional Techniques: `scope.techniques` — default("Standard AI simulation, benign telemetry generation, and lab-scoped scanning techniques.")

Limitations: results are representative for the test environment and detectors used; real production systems and defensive stacks may behave differently. Model performance depends on the quality of telemetry and labelled data available.

Recon Findings

Host Information

- Operating System: `recon_data.os_name``recon_data.os_version``Architecture : recon_data.architecture`
- Patch Level / Updates:
- Installed Software:
 - sw
 - N/A

User & Account Details

- Privileged Accounts:
 - acc
 - None
- Active Users:
 - user
 - None
- Default Credentials Detected: `recon_data.default_credentials|default("None")`

Network Topology

- IP Addresses:
- Subnets:
- Firewalls:
- VPN / DMZ / Segmentation Info: `recon_data.vpn_dmz|default("N/A")`

Open Ports & Services

- Port `port.number` / `port.protocol`
- None

Active Processes & Services

- `proc.name` (PID: `proc.pid` , User: `proc.user`)
- N/A

Endpoints & Devices

- dev.type — dev.identifier (Status: dev.status)
- None

Active AV / EDR

- av.name — Version: av.version — default("Unknown")
- None detected

Misconfigurations Detected

- misconf
- None

Findings & Analysis

This section analyzes the data collected by the Red Team AI Malware Simulator, highlights critical issues, and provides context for the AI’s actions and attack outcomes. It focuses on risk assessment and patterns identified during the simulation.

Finding / Issue	Evidence / Data	Severity	Potential Impact if System Were Real
f.name — default("N/A")	f.evidence — default("N/A")	f.severity — default("Medium")	f.impact — default("N/A")

Attack Simulation Summary

Purpose: This section summarizes the actions executed by the simulated malware, including their outcomes and prioritization. It provides an overview of what was attempted, what succeeded or failed, and how the AI prioritized each action.

Actions Executed: The table below lists all simulated attack actions performed by the Red Team AI Malware Simulator.

Action Name	Description
a.name — default("N/A")	a.description — default("N/A")
No actions executed	N/A

LLM Based Mitigations

Purpose: This section uses a pre-trained Large Language Model (LLM) to analyze the collected simulation data and generate recommended mitigations for the findings. It provides a concise “briefing for action” rather than a deep technical manual.

Note on LLM Limitations:

- Recommendations are based solely on the simulation data provided.
- The LLM may not capture every nuance of real-world system configurations or risks.
- Human verification and contextual analysis are recommended before applying any mitigations.

Recommended Mitigations:

- m — *latex_escape*
- No mitigation suggestions available.

Raw Data

```
{{ raw_data | safe }}
```